ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE
PRESENTS

# CONTRAPTIONS

Offical Event Rulebook

AVISHKAR '26
WHERE CURIOSITY BECOMES CREATION

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1.**Theme: Energy Conversion – Hit & Drop Challenge:** This event focuses on creating complex chain reactions that demonstrate energy transformations to hit and move a bottle.

2.**Event Description:** A contraption involves turning simple tasks into creative, complex chain reactions where teams design systems showing multiple energy conversions with precision and robustness.

3.**Problem Statement:** Participants must design a multi-step contraption that uses various energy conversions to trigger a sequence that eventually hits and displaces a bottle.

4.**Task Flow:** The sequence must start with an initial trigger, followed by energy conversions, hitting the bottle, and finally dropping the bottle at least 1 foot away from its original spot.

5.**Energy Conversion Requirement:** Systems must use at least 3 energy conversions (such as mechanical, electrical, elastic, or magnetic), and each step must automatically trigger the next.

6.**Team Details:** Each team must consist of exactly 4 mandatory participants.

7.**Arena Constraints:** The maximum allowed size for the contraption setup is 2 meters by 4 meters.

8.**Time Rules:** Teams are given a total of 15 minutes and are allowed a maximum of 3 trials.

9.**Intervention Rules:** A maximum of 3 human interventions are permitted; manual interference between steps is not allowed, and extra interventions result in penalties.

10.**Task Specifications (HIT & FINAL DROP CONDITION):** The bottle must be struck by a moving mechanism (no direct hand interaction) and must fall and land with clear displacement of at least 1.5 feet (≈30 cm).

11.**Materials Rules:** Participants must bring all their own materials; no materials are provided by organizers, no sharing is allowed, and the use of waste materials is encouraged.

12.**Safety Rules:** Designs must be safe and stable, using no hazardous materials, and teams must clean their area after they finish.

13.**Judging Criteria:** Teams are evaluated on the number of steps, energy conversions, innovation, successful hit/drop (mandatory), execution smoothness, robustness, and bonuses for minimal intervention or waste material use.

**14.Penalties:** Deductions are given for manual interventions and failing the 1.5-feet drop, while unsafe designs result in disqualification.

15.**Eligibility:** The competition is open to all undergraduate (UG) students with a valid ID card.

16.**Example Flow:** A sample sequence includes a raised ball falling (potential to kinetic), hitting a lever (mechanical), releasing a spring (elastic to kinetic), hitting the bottle, and finally throwing the bottle 1 foot away.

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